

IB DP Mathematics AA HL

AHL 5.18 — First-Order Differential Equations

Exam-Style Practice Question Bank

10 questions of each specified type • 40 questions total

Topic	Questions
Euler's method	1–10
Variables separable	11–20
Homogeneous DEs, $y = vx$	21–30
Linear DEs, integrating factor	31–40

Instructions: Show all necessary working. Give exact answers where appropriate. For numerical answers, use the requested accuracy. Questions are written in an IB-style format but are original practice questions, not past-paper questions.

A. Euler's Method — Numerical Solutions

1. Using Euler's method with step size $h = 0.2$, estimate $y(0.6)$ for $dy/dx = x + y$, given $y(0) = 1$.
2. Use Euler's method with $h = 0.1$ to estimate $y(0.3)$ for $dy/dx = y - x^2$, given $y(0) = 1$.
3. Starting from $(0, 2)$, use Euler's method with $h = 0.25$ to obtain an estimate of $y(1)$ for $dy/dx = xy$.
4. For $dy/dx = \sin x + y$, $y(0) = 0$, use Euler's method with $h = 0.2$ to estimate $y(0.8)$.
5. The differential equation is $dy/dx = x/y$, with $y(1) = 2$. Use Euler's method with $h = 0.1$ to estimate $y(1.4)$.
6. Given $dy/dx = y(1 - x)$, $y(0) = 2$, use Euler's method with $h = 0.2$ to estimate $y(1)$.
7. A numerical solution is required for $dy/dx = x^2 - 2y$, $y(0) = 1$. Use Euler's method with $h = 0.1$ to estimate $y(0.5)$.
8. For $dy/dx = (x + y)/2$, $y(0) = 3$, use Euler's method with $h = 0.25$ to estimate $y(0.75)$.
9. Use Euler's method with $h = 0.2$ for $dy/dx = e^{-x} + y/2$, $y(0) = 1$, and obtain an estimate of $y(0.6)$.
10. A model satisfies $dy/dx = y \cos x$, $y(0) = 1$. Use Euler's method with $h = \pi/10$ to estimate $y(\pi/2)$. Give your answer to 3 significant figures.

B. Variables Separable

11. Solve $dy/dx = 3x^2y$, given $y(0) = 2$.
12. Solve $dy/dx = (x + 1)(y + 2)$, given $y(0) = 1$.
13. Solve $dy/dx = x/(1 + y^2)$, given $y(0) = 0$.
14. Solve $dy/dx = y(1 - y)$, given $y(0) = 1/2$.
15. Solve $dy/dx = (2x)/(y)$, given $y(1) = 3$.
16. Solve $dy/dx = y/x$, given $y(2) = 5$.
17. Solve $dy/dx = x e^{(-y)}$, given $y(0) = 0$.
18. Solve $dy/dx = (1 + x^2)/(1 + y^2)$, given $y(0) = 1$.
19. A population P satisfies $dP/dt = 0.04P(1 - P/5000)$. Given $P(0) = 1000$, find $P(t)$.
20. A particle satisfies $dv/dt = -kv^2$, where $k > 0$. Given $v(0) = v_0$, derive an expression for v in terms of t , k and v_0 .

C. Homogeneous Differential Equations — substitution $y = vx$

- 21.** Solve $dy/dx = (x + y)/x$ using the substitution $y = vx$.
- 22.** Solve $dy/dx = (y/x)^2 + y/x$ using $y = vx$.
- 23.** Solve $dy/dx = (x^2 + y^2)/(xy)$ using the substitution $y = vx$.
- 24.** Solve $dy/dx = 1 + y/x$ using $y = vx$, and find the solution satisfying $y(1) = 2$.
- 25.** Solve $dy/dx = (x - y)/(x + y)$ using $y = vx$.
- 26.** Solve $dy/dx = (x^2 - y^2)/(2xy)$ using the substitution $y = vx$.
- 27.** Solve $dy/dx = (y^2 + xy)/x^2$ using $y = vx$.
- 28.** Solve $dy/dx = (x + 2y)/(2x + y)$ using the substitution $y = vx$.
- 29.** Solve $dy/dx = (y/x) + (x/y)$ using $y = vx$, and state any restrictions on the solution.
- 30.** Solve $dy/dx = (2x^2 + y^2)/(xy)$ using $y = vx$, given $y(1) = 1$.

D. Linear First-Order Differential Equations — Integrating Factor

31. Solve $y' + 2y = e^x$.

32. Solve $y' - 3y = x^2$, given $y(0) = 1$.

33. Solve $y' + (1/x)y = x^2$, for $x > 0$.

34. Solve $y' + 2xy = 4x$, given $y(0) = 1$.

35. Solve $y' - (2/x)y = x^3$, for $x > 0$.

36. Solve $y' + y \tan x = \sin x$, on an interval where $\cos x \neq 0$.

37. Solve $y' + (3/x)y = 6x^2$, given $y(1) = 2$.

38. Solve $y' - 4y = 2e^{4x}$, given $y(0) = 3$.

39. Solve $y' + (2/(x + 1))y = x$, for $x > -1$.

40. A temperature T satisfies $T' + 0.1T = 2$, with $T(0) = 20$. Find $T(t)$, and hence determine the limiting temperature as $t \rightarrow \infty$.